

TFPT verification suite: the journey of ~475 machine-checked scripts

v1-v23 - Foundations

carrier $D5+A3+\mu_4$, E8 glue, α^{-1} fixed point,
anchor $a=(1,1,2) \rightarrow \{c3,g_car\}$ reduce to $\{a,\pi\}$

v24-v53 - Standard-Model readouts

ϕ_0 mass ladder, flavor (Q,K,R,L), lepton/quark
ratios, hypercharge, compiler core (5,3)

v54-v100 - Seam = horizon

one-sided Gauss-Bonnet $c3=1/(8\pi)$, Coxeter-30 cycle,
gapped attractor $(2/3)^6$, frozen registry + null-MC

v101-v140 - Horizon + flavor geometry

Nariai=anchor, monodromy= $W(A3)=S4$, cusp
weights $\{0,1/3,2/3\}$, H^1 cohomology, canonical map

v141-v158 - R1-R5 + premise (A)

deck selection, EH mechanism, No-Unit Theorem,
simple-current (E8) $_1$, free $c=8$ fixed point isolated/stable

v159-v169 - PyR@TE cross-checks

SM gauge/Yukawa/Higgs RGEs confirmed ($b1=41/10$),
 Λ_{QCD} , m_H near-criticality, η_B leptogenesis

v170-v174 - AQFT bridges

E8 slice compression, OS moment/Sugawara, trace
anomaly seed $4/3$, strong-CP Pfaffian, Fock readings

v175-v181 - AQFT closure -> geometric bedrock

net existence + full-cone RP [E];
residual -> one premise: carrier μ_4 clock = seam conformal deck

v182-v213 - F_transfer functor + frontier

reviewer residual map; $F_transfer$ = one typed
functor (Koide, η_B , axion, m_p/m_e) + machine guard

v214-v218 - Pillowcase + Sheet Diamond

QGEO pillowcase (cross-ratio $2 \Rightarrow j=1728$), four
marks from Gauss-Bonnet, the Sheet Diamond axis geometry

v219-v237 - Icosahedral capstone

McKay why $\{2,3,5\}$, CM norms $41/7$, CP triality, Kleinian
seam, $\det K=1$ = Kitaev E8; the (2,3,5) Brieskorn generates ALL

v238-v261 - NCG / Modular Spectral Closure

96-dim spectral triple, Dirac = covariance induction,
cutoff = KMS weight, seam/carrier/E8 on one $K3$: one relative object

v262-v275 - Frontier closure + QFT4D fork

F_QCD m_p/m_e , M_ν seesaw, S_pert (EG, 1-loop + gauge
betas), scale over-determination, the QG.AMB roadmap

v276-v299 - The Gral: SEAM.EQUIV.01 + Flat-Away

seam = (E8) $_1$ at $\tau=i$ named as ONE theorem;
both routes reduced to one shared input (Flat-Away), heat $a2$ proved+Lean (v295/v296)

v300-v302 - Closing arc: residual pinned

Flat-Away hard+pin from (E8) $_1$ (v300); Route A invertible
via free fermions (v301); gap = derived $6\ln(3/2)>0$ (v302); no TFPT-internal assumption left

v303-v407 - Solvers + parameter-free gravity + closure

typed $F_transfer$ solvers (Koide/ η_B / m_p/m_e /axion, v371-v375/v402);
parameter-free Einstein eq. full nonlinear (v359); QG.AMB a [C] redundancy (v369)

v408-v481 - Certification + external bridges

α -Quillen narrowed to the zeta-det face (inflow $|C|=1$, det-line moduli, v470/v472);
entropic-action bridge typed (v473-v478); Kronheimer v479; four-interval μ_4 v480; nu-scale v481